

# Accelerating metabolic engineering through multimodal phenotype prediction with TorchCell and the Cell Graph Transformer

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## Abstract ( $\leq 150$ words)

Deep learning has advanced protein structure prediction, yet metabolic engineering lags because phenotype data are scattered across heterogeneous studies lacking a shared ontology. We present TorchCell, an open-source framework that annotates literature and public datasets with a shared ontology and ingests them into a Neo4j knowledge graph, then builds deep-learning-ready datasets in which each strain is a perturbation operator over a shared wildtype reference cell combining genome sequence, gene networks, metabolism, and environment. We develop the Cell Graph Transformer (CGT), a virtual-cell architecture in which biological interaction graphs constrain multi-head attention and an equivariant perturbation operator feeds multitask decoders. In *Saccharomyces cerevisiae*, CGT predicts trigenic gene-gene interactions (Pearson  $r = 0.454$ ), outperforming DANGO, DCell, and Yeast9 flux balance analysis, and generalizes to knockout expression ( $r = 0.543$ ) and morphology ( $r = 0.619$ ). CGT recommends gene deletions for beta-carotene and betaxanthin production, pairing with autonomous biofoundries for AI-guided strain engineering.

**Keywords:** metabolic engineering, foundation models, gene interactions, virtual cell, knowledge graph, strain design

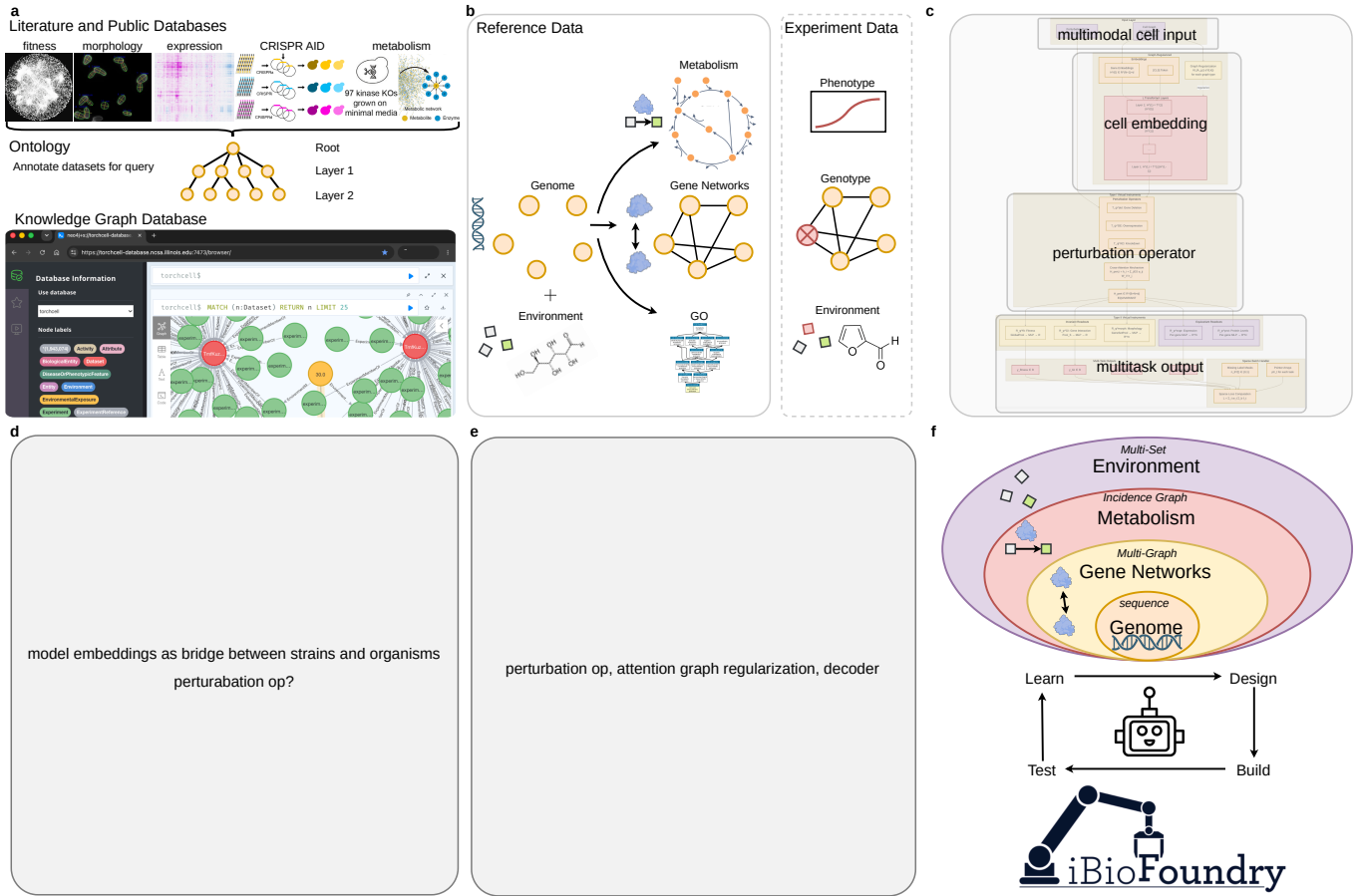
## 1 Introduction (target 570 words)

[FILLER – replace with Intro prose.] Deep learning transformed protein structure and function prediction, yet systems biology and metabolic engineering lag because phenotype data are scattered across small, heterogeneous studies with no shared schema [1]. Prior approaches fall short: mechanistic flux balance analysis misses epistasis, and recent perturbation transformers do not consistently beat linear baselines [2]. Our own classical baselines predict knockout fitness but fail on gene interactions (Suppl. Fig. S1), motivating a deep, biologically grounded model. Here we present TorchCell and the Cell Graph Transformer (CGT), whose data model and architecture we outline in Suppl. Fig. S7.

## 2 Results (target 1,900 words)

**A shared ontology and knowledge graph unify metabolic-engineering data (380 words).** [FILLER – R1.] TorchCell annotates literature and public datasets with a shared experiment ontology and ingests them into a Neo4j knowledge graph; dataloaders emit deep-learning-ready datasets in which each strain is a perturbation operator over a shared wildtype reference cell (Fig. 1; data scale in Suppl. Fig. S2; ontology/KG spec in Suppl. Fig. S4).

**The Cell Graph Transformer predicts trigenic gene-gene interactions at state of the art (380 words).** [FILLER – R2.] The Cell Graph Transformer (CGT; architecture in Fig. 1c and Methods) constrains attention with biological interaction graphs and maps the reference cell through an equivariant perturbation operator. On trigenic gene-gene interactions (GGI), where classical ML fails (Suppl. Fig. S1), CGT reaches Pearson  $r = 0.454$  (Spearman 0.421), outperforming Yeast9 FBA, DCell, and DANGO (Fig. 2).



**Fig. 1** TorchCell: a database and software framework for building deep-learning-ready metabolic-engineering datasets. (a) Literature and public databases are annotated with TorchCell's experiment ontology and ingested into a Neo4j knowledge graph (hosted at NCSA), queryable across datasets. (b) Genomic representation: reference data (genome, gene networks, metabolism, GO) and experiment data (phenotype, genotype, environment); experiments are compressed into perturbation operators over a shared reference state. (c) The Cell Graph Transformer (CGT) maps multimodal cell input to a cell embedding via a perturbation operator, decoded into multitask outputs. (d) Model embeddings bridge strains and organisms. (e) Perturbation operator, attention graph regularization, and decoder. (f) The multimodal cell representation feeds a Learn–Design–Build–Test loop with the NSF iBioFoundry.

[Fig. 2 placeholder – compose in draw.io]

Trigenic gene-gene interactions. (a) concept — why trigenic interactions are hard to predict; (b) predicted (CGT) vs. actual (experimental) interaction score; (c) state of the art vs. Yeast9 FBA / DCell / DANGO; (d) accuracy stratified by interaction magnitude  $|\tau|$ ; (e) data and model scaling ( $r$  vs. #interactions and model size); (f) attention-graph-regularization ablation (does graph-constrained attention add value?).  
*Per plan: can split into two figures if crowded.*

**Fig. 2** State-of-the-art trigenic gene-gene interaction prediction. (a)–(f).

**One embedding, many phenotypes: expression, morphology, and fitness** ( 380 words). [FILLER – R3.] The same embedding predicts knockout expression ( $r = 0.543$ ), single-knockout morphology ( $r = 0.619$ ), and fitness (Fig. 3; expression diagnostics in Suppl. Fig. S5).

**Natural genetic variation versus model-designed perturbations** ( 380 words). [FILLER – R4.] We compare phenotypes from natural strain genetic variation against deep-learning-designed gene perturbations across genome and environment (Fig. 4).

[Fig. 3 placeholder – compose in draw.io]

Expression, morphology, and fitness. (a) expression predicted vs. actual (top genes); (b) morphology predicted vs. actual; (c) fitness predicted vs. actual; (d) cross-phenotype summary / shared embedding; (e) trends across phenotype classes.

**Fig. 3** One latent cell embedding generalizes across expression, morphology, and fitness. (a)–(e).

[Fig. 4 placeholder – compose in draw.io; from handwritten plan, confirm panels]

Natural variation vs. DL design. (a) natural strain genome variation and phenotype; (b) DL-predicted phenotype across the genome  $\times$  environment landscape; (c) model-designed knockouts compared with natural variants.

**Fig. 4** Natural genetic variation versus model-designed perturbations. (a)–(c).

[Fig. 5 placeholder – compose in draw.io; from handwritten plan, confirm panels]

Metabolism. (a) metabolic-network grounding / additional models; (b) benchmark; (c) kinase-knockout collection; (d) isobutanol; (e) tryptophan-pathway precursors; (f) beta-carotene / betaxanthin production with the iBioFoundry DBTL loop.

**Fig. 5** Extending TorchCell to metabolism and metabolic-engineering strain design. (a)–(f).

**Extending to metabolism and strain design** ( 380 words). *[FILLER – R5.]* Grounding the cell representation in metabolism, CGT recommends gene targets for production phenotypes and pairs with the UIUC iBioFoundry for iterative design (Fig. 5; inference at scale in Suppl. Fig. S6).

### 3 Discussion (target 570 words)

*[FILLER – Discussion.]* A strain-aware, biologically grounded representation is the differentiator; limitations include single-organism scope, morphology-data maturity, and error-bar provenance.

## 4 Methods (no limit)

### Ontology and knowledge-graph construction

[FILLER.] Experiment ontology (genotype, phenotype, environment, experiment metadata); annotation of literature and public datasets; ingestion into the Neo4j knowledge graph; conversion, deduplication, and aggregation into HeteroData records (Suppl. Figs. S8, S9).

### Datasets

[FILLER.] Source datasets and versions (e.g. Costanzo, Kuzmin trigenic; knockout expression; single-knockout morphology); inclusion/exclusion criteria; dataset sizes and label provenance.

### Cell representation and the perturbation operator

[FILLER.] Shared wildtype reference cell; each strain as a perturbation operator over the reference; multimodal node/edge features (genome sequence, gene-interaction networks, metabolism, environment); DNA sequence selection and tokenization (Suppl. Figs. S10, S11).

### Data splits and preprocessing

[FILLER.] Train/validation/test partitioning (and any gene- or interaction-disjoint splits); label normalization; handling of missing labels.

### Cell Graph Transformer architecture

[FILLER.] Graph-constrained multi-head attention; the equivariant perturbation operator mapping reference embedding to perturbed state; multitask decoder heads (Fig. 1c). Layer counts, hidden dimensions, and parameter budget.

### Graph-regularized attention and training objective

[FILLER.] Attention-prior alignment loss aligning attention to network priors; multitask loss weighting; full mathematical formulation.

### Model training

[FILLER.] Optimizer, learning-rate schedule, batch size, early stopping; hardware (GPUs, distributed/DDP), wall-clock; software versions; random seeds.

### Baselines

[FILLER.] DANGO, DCell, and Yeast9 flux balance analysis; classical-ML baselines (Suppl. Fig. S1); matched training data and evaluation protocol.

### Evaluation metrics

[FILLER.] Pearson correlation ( $r$ ) and any secondary metrics; aggregation across folds; definition of error bars (see Statistics).

### Gene-gene interaction prediction

[FILLER.] Trigenic interaction prediction setup; performance vs. interaction magnitude; ablations (Fig. 2).

### Expression and morphology prediction

[FILLER.] Knockout transcriptome and single-knockout morphology tasks; shared latent embedding; cross-phenotype transfer (Fig. 3).

### CGT-guided strain design

[FILLER.] Objective and candidate enumeration for beta-carotene and betaxanthin production; ranking and selection of recommended gene deletions (Fig. 5).

## Strain construction and cultivation

[*FILLER.*] Strains, plasmids, and genome edits; the TorchCell + UIUC iBioFoundry design–build–test–learn loop; cultivation and fermentation conditions.

## Metabolite quantification

[*FILLER.*] Sampling, extraction, and analytical method (HPLC/LC–MS); calibration and titer quantification.

## Inference at scale

[*FILLER.*] Large-scale inference pipeline and resources (Suppl. Fig. S6).

## Statistics

[*FILLER.*] Statistical tests used and whether one- or two-sided; sample sizes ( $n$ ); definition of center and dispersion; box-plot conventions (median, IQR hinges, whiskers  $\leq 1.5 \times \text{IQR}$ ); multiple-comparison handling; significance thresholds.

## Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

## Data availability

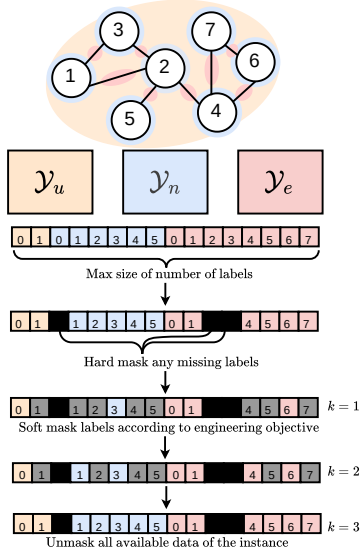
[*FILLER.*] Datasets are available on Hugging Face (<https://huggingface.co/...>); the Neo4j knowledge graph is hosted at NCSA (...); trained model weights are available at .... Source data are provided with this paper.

## Code availability

[*FILLER.*] TorchCell and the Cell Graph Transformer are open source at <https://github.com/Mjvolk3/torchcell> (license: ...), with a release archived at Zenodo (<https://doi.org/...>).

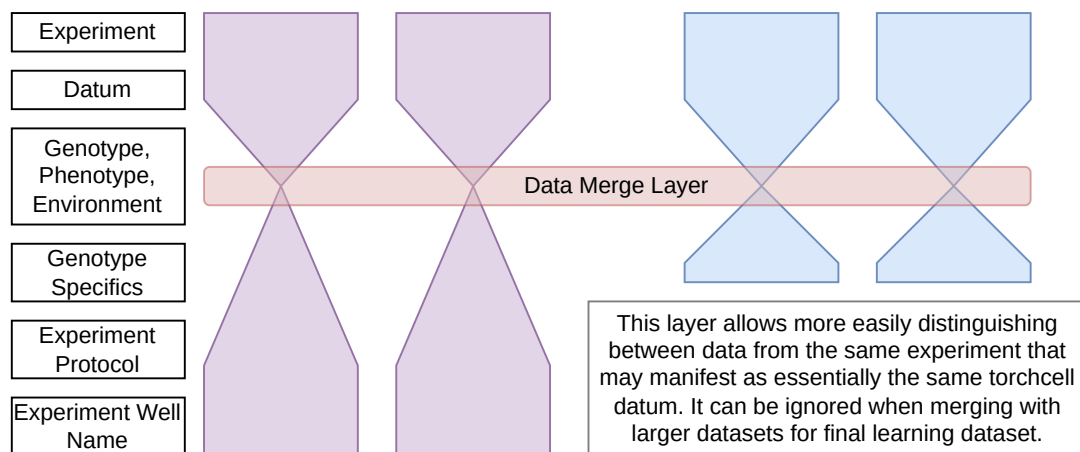
## Acknowledgments.

## Declarations

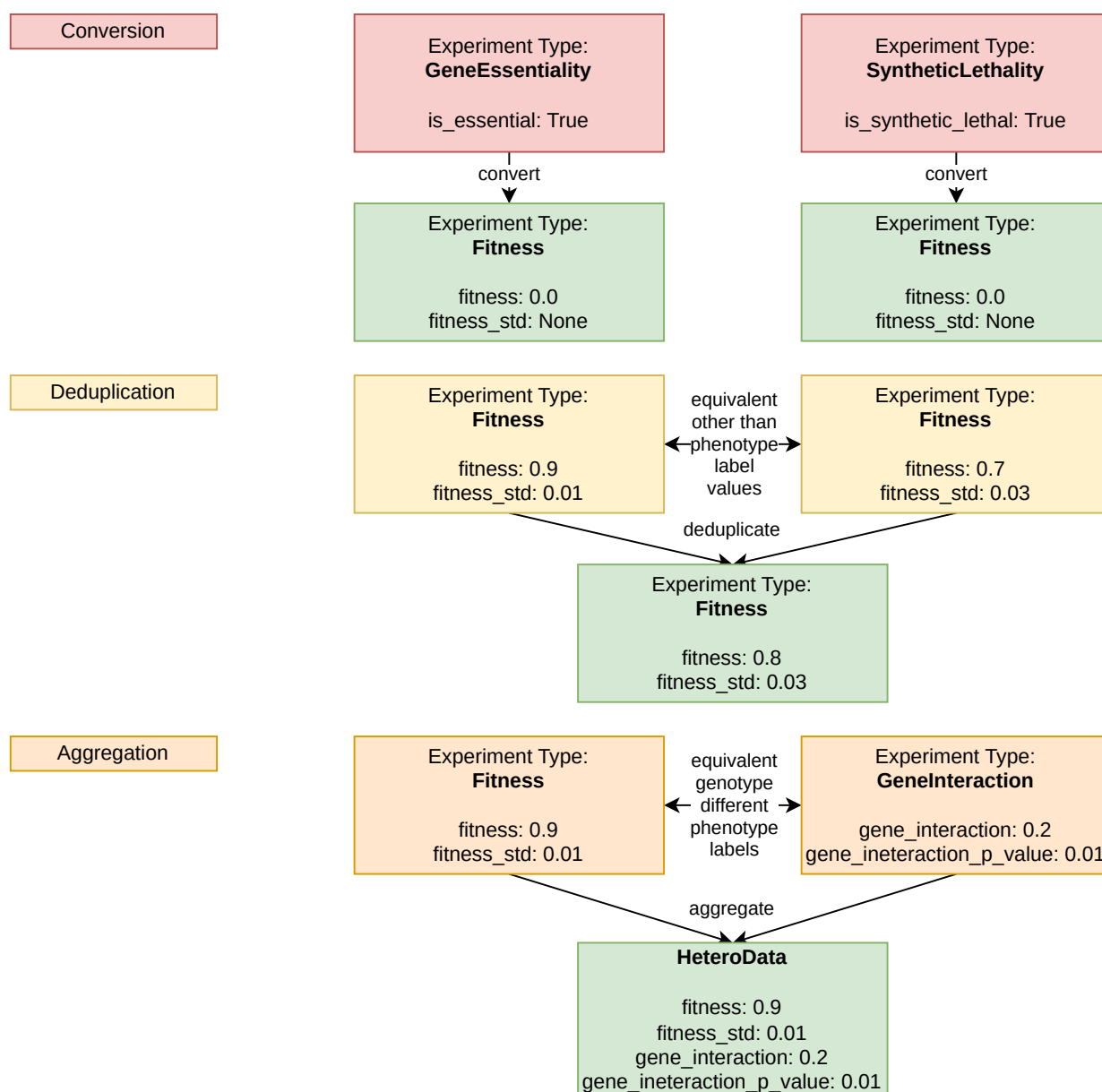


**Fig. S7** TorchCell: ontology-unified knowledge graph and the perturbation-operator cell representation. Panels (a)–(d).

**Supplementary information.**



**Fig. S8** Ontology data model (hourglass). Each experiment funnels genotype, phenotype, environment, and experiment-specific metadata through a shared Data Merge Layer that distinguishes records from the same experiment; the layer can be ignored when merging into larger datasets.



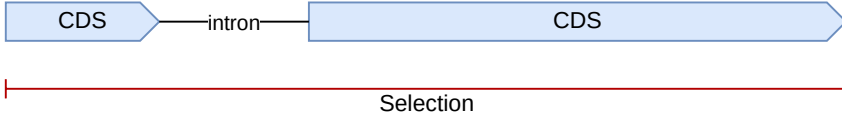
**Fig. S9** Knowledge-graph normalization in three stages. Conversion maps phenotype experiment types (GeneEssentiality, SyntheticLethality) to a common Fitness type; deduplication collapses records equivalent other than their phenotype-label values; aggregation combines equivalent-genotype records with different phenotype labels (Fitness, GeneInteraction) into a single HeteroData instance.

## DNA Selection

To comply with the standard format using in DNA language models we want to select the DNA sequence that has the highest probability of having the outermost start and stop codons for the CDS of the the gene. This means we will ignore five prime UTR introns if they are not internal to the gene. Below we outline how we handle intron cases:

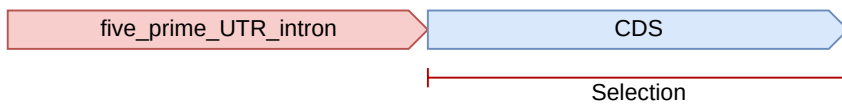
### YFL039C - Internal Intron

Selectin - Entire gene



### YBL072C - Upstream five prime UTR intron

Selection - Gene

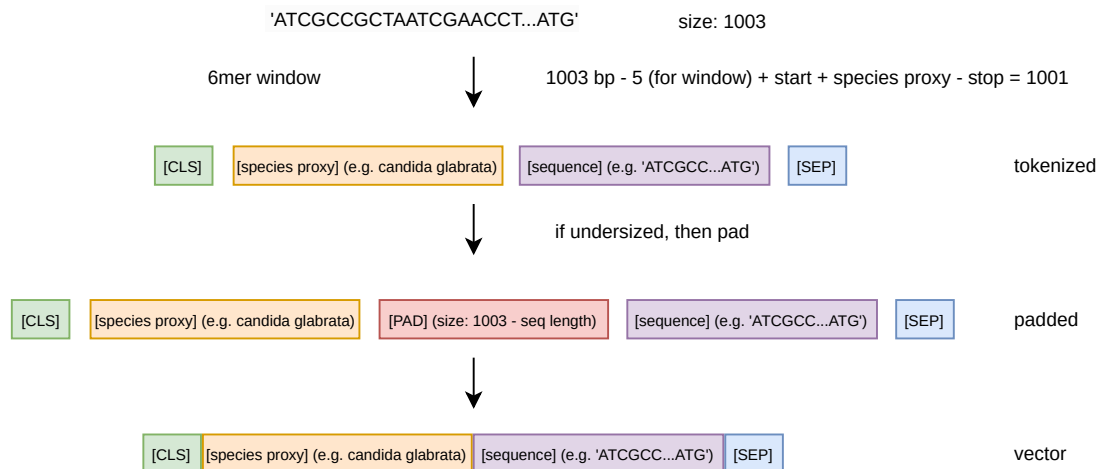


### YIL111W - 1bp CDS

Selection - Gene (The only case with no start codon)

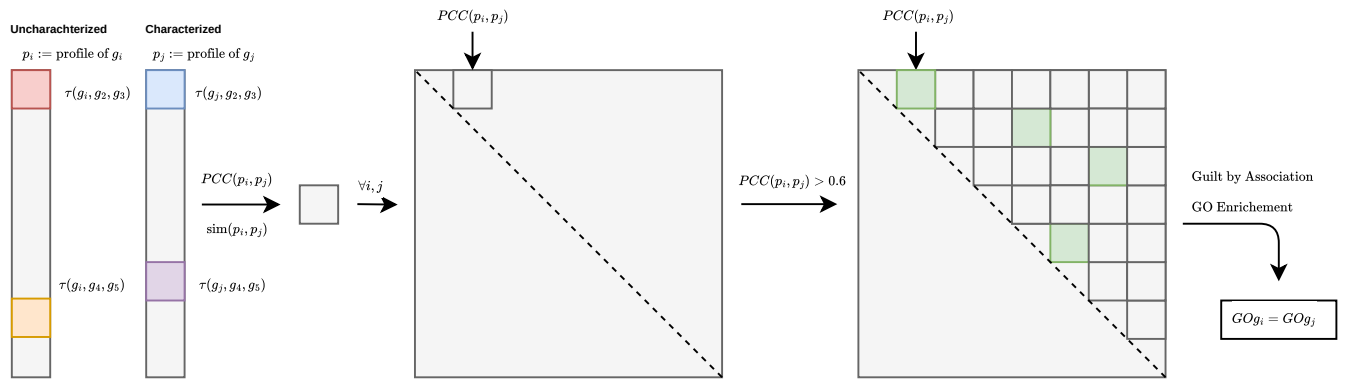


**Fig. S10** DNA sequence selection for the sequence model. The coding sequence is taken from the outermost start to stop codon, ignoring five-prime UTR introns unless internal to the gene. Cases: internal intron (YFL039C), 5' UTR intron (YBL072C), and a 1 bp CDS lacking a start codon (YIL111W).



**Fig. S11** DNA tokenization. A 1003 bp window is encoded as [CLS], species proxy, 6-mer sequence, [SEP]; undersized sequences are padded with [PAD] to a fixed length before vectorization.





**Fig. S12** Guilt-by-association annotation of uncharacterized genes. (a)–(e) Triple-interaction profiles  $p$  are correlated (Pearson, PCC) across all gene pairs; pairs with  $PCC > 0.6$  are linked, and GO enrichment over the resulting network transfers annotations ( $GO_{g_i} = GO_{g_j}$ ). (f) Total quantification of triple interactions across the remaining yeast genome (coverage of uncharacterized genes). (g) The resulting new annotations transferred to previously uncharacterized genes.

## References

- [1] Volk, M. J., Tran, V. G., Tan, S.-I., others & Zhao, H. Metabolic engineering: Methodologies and applications. *Chemical Reviews* **123**, 5521–5570 (2023).
- [2] Ahlmann-Eltze, C., Huber, W. & Anders, S. Deep-learning-based gene perturbation effect prediction does not yet outperform simple linear baselines. *Nature Methods* **22**, 1657–1661 (2025).
- [3] Ma, J. *et al.* Using deep learning to model the hierarchical structure and function of a cell. *Nature Methods* **15**, 290–298 (2018).
- [4] Zhang, R., Ma, J. & Ma, J. Dango: Predicting higher-order genetic interactions. *bioRxiv* (2020).
- [5] Merzbacher, C., Mac Aodha, O. & Oyarzún, D. A. Accurate prediction of gene deletion phenotypes with flux cone learning. *Nature Communications* **16**, 8492 (2025).
- [6] Bunne, C. *et al.* How to build the virtual cell with artificial intelligence: Priorities and opportunities. *Cell* **187**, 7045–7063 (2024).